

1(a)	work done per unit mass	M1
	bringing (small test) mass from infinity (to the point)	A1
1(b)(i)	$\Delta\phi = (GM/2R) - (GM/5R) = 3GM/10R$	A1
1(b)(ii)	change in GPE = $(3 \times 4.0 \times 10^{14} / 10 R) \times 4.7 \times 10^4$	C1
	$(3 \times 4.0 \times 10^{14} / 10 R) \times 4.7 \times 10^4 = (1.70 - 0.88) \times 10^{12}$ $R = 6.88 \times 10^6$	C1
	distance = $3 \times 6.88 \times 10^6$ = 2.1×10^7 m	A1

Question	Answer	Marks
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